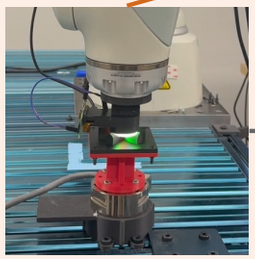
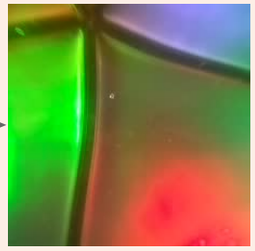
## (a) Register real contacts

*Labels need calibration and verification*

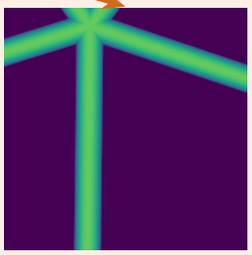
$q^*$ from the robot pose



Registered press

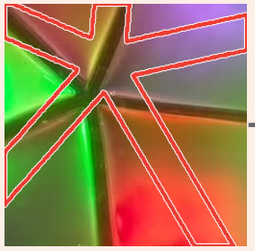


Real image



**GT label**

after re-placing the object



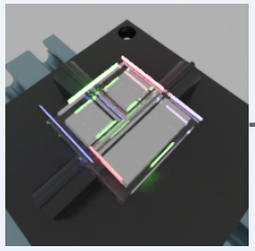
old calibration



re-calibrated, verified

## (b) Simulate contacts

*Synthetic RGB needs sensor-response modeling*

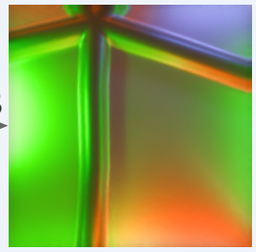


Simulated contact

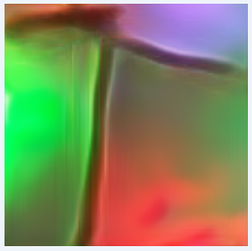


**GT label** (free)

render RGB

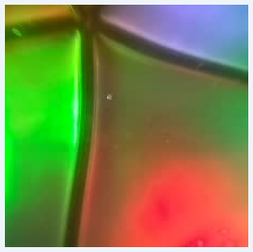


PBR



Learned
(fitted on real pairs)

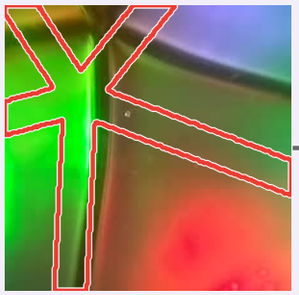
$\neq$



Real image

## (c) AlignTac: real appearance + aligned geometry

unlabeled real image + known mesh



mesh at teacher estimate

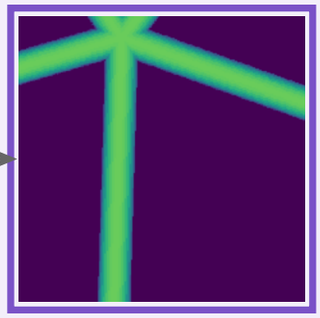
**Learned aligner $\mathcal{A}$**

train

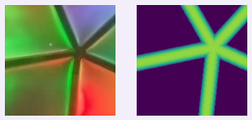
$q \leftarrow q + \Delta q$



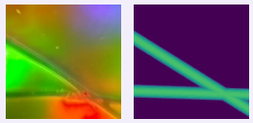
mesh after alignment



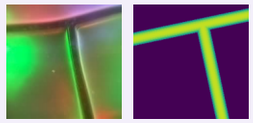
**Pseudo-label**



*K* **registered contacts**







pseudo-labels for the remaining real images

$K$=25 with AlignTac ≈
real-only $K$=50
(contact metrics)